Bruno Saboia de Albuquerque

e-mail: bruno@saboia.it – Tel: +41 78 245 7688
GitHub – *LinkedIn*

Summary

Senior software engineer with over 20 years of experience delivering production systems for banking, insurance and international organisations — most recently conversational AI at AXA Group Operations and data analytics at the Bank for International Settlements. Works across the full lifecycle: C#, Python, SQL and TypeScript; DevOps, CI/CD and Kubernetes; and applied AI/ML, LLM and RAG systems. Economics graduate, published author and peer reviewer, experienced in distributed, multilingual teams across Europe and the Americas.

Experience

Conversational AI Engineer *Oct 2025 – Jul 2026 (9 months)*
AXA Group Operations, Winterthur, CH
— *https://axa.com*

Built the Python API and React front end of the LLM assistant used by AXA's claims handlers in Switzerland, under PII redaction, GDPR and cybersecurity constraints.

- Reduced claim handling time by 30% for the handlers using the assistant
- Rolled out to handlers covering 50% of AXA Switzerland's claim volume
- Owned the internal developer tooling the rest of the chatbot team built on
- Took an almost-greenfield prototype to production readiness for rollout into France and Spain

Technologies: TypeScript, React, Python, LLM, RAG, Azure, Docker, GitHub

Data Analytics Engineer *May 2024 – May 2025 (1 year)*
Bank for International Settlements, Basel, CH
— *https://bis.org*

Built BIS GPT, the bank's internal LLM assistant, as one of its core engineers, covering the Python API, the React front end, the OpenShift/ArgoCD delivery pipeline, and the internal tooling the other engineers used.

- Achieved bank-wide adoption: BIS GPT is used across the entire institution
- Owned the project's Azure stack end to end, running on private OpenAI endpoints so no data left the bank's control
- Enforced the bank's document classification along the retrieval path, so answers respected its information-handling rules
- Contributed to Voyager, the bank-wide AI programme, delivering jurisdiction extraction for Procurement in place of manually reading the bank's international contracts, on the judgement that the use case tolerated approximate accuracy

Technologies: TypeScript, React, Python, LLM, RAG, Azure, OpenShift, Docker, ArgoCD, GitHub, Microsoft Foundry

Business Analyst/Developer *May 2022 – May 2024 (2 years)*
Bank for International Settlements, Basel, CH
— *https://bis.org*

Built and modernised the systems Risk Management used to measure the bank's credit, market, liquidity and counterparty risk, gathering requirements directly from risk analysts and implementing them end to end, and containerised the department's applications onto OpenShift.

- Migrated 100% of the department's pipelines from Azure DevOps to GitHub
- Designed, built and solely owned a project-template CLI, adopted for 100% of new projects

- Cut the CREC (Credit Risk Economic Capital) run from roughly four hours to 30 minutes by making a CUDA implementation the default, keeping the C# CPU path as fallback
- Delivered Credit Review, a WPF tool for the risk analysts' daily credit review operations

Technologies: C#, Git, Azure DevOps, Python, PowerShell, Bash, OpenShift, SQL, Octopus, ArgoCD

Senior Software Engineer *Mar 2022 – May 2022 (2 months)*
Clevertech, Remote (Brazil – US)
— *https://clevertech.biz*

Clevertech is an American IT company, providing consulting services for large customers around the globe.
Built the Postgres and Airflow pipelines behind a client's reporting dashboards, delivering the contract to completion.

Technologies: Python, Apache Airflow, Postgres

Full-stack developer *Sep 2021 – Mar 2022 (6 months)*
Tesseract, Remote (Brazil – Finland)
— *https://tesseractinvestment.com/*

Tesseract is a Finnish startup which borrows cryptocurrencies, matchmaking investors and lenders. It is Fin-FSA regulated digital asset innovator, backed by reputable investors, focusing on institutional brokerage and asset management.
Established the company's first Infrastructure-as-Code platform on Pulumi and GitHub Actions for a greenfield build, replacing ClickOps provisioning with reproducible, versioned infrastructure.

Technologies: Pulumi, TypeScript, GitHub Actions

Senior Software Engineer *Apr 2021 – Sep 2021 (5 months)*
BEC Financial Technologies, Roskilde, Denmark
— *https://bec.dk*

BEC is a Danish fintech that provides services for medium-to-small banks. Founded in 1964, is one of the main back-bones of IT services for financial companies in the Nordic country.
Ran Kafka as an internal platform for the teams building BEC's payment systems, the event-streaming backbone serving small and medium-sized Danish banks. Introduced GitOps for topics and other Kafka objects, and made incident fixes reproducible and auditable by scripting them onto their tickets, so the change proven in lower environments was the one applied in production.

Technologies: Kafka, Java, Bash, OpenShift, Git, CDC, Docker, Jenkins

Senior Software Engineer *Feb 2019 – Mar 2021 (2 years, 1 month)*
Danske Bank, Copenhagen, Denmark
— *https://danskebank.dk*

Built SmartSearch, the corporate payment search product of Danske Bank's Payment Innovations department: change data capture off the DB2 payments table into Kafka Streams, indexed into Elasticsearch, keeping the proven COBOL/DB2 stack as the source of truth. Wrote the C# API behind its Angular UI, reconciled payments across both stores, and moved the stack onto Kubernetes.

- Cut corporate payment search from over three minutes to return around 3,000 payments to a few milliseconds returning 500,000
- Took the project through its relocation from Copenhagen to Aarhus during a hiring freeze, having joined as a senior developer

- Migrated most of the Kafka Streams codebase from Java to Kotlin, starting with the test suite, cutting verbosity at no performance cost on the JVM
- Rewrote the PDF report microservice in Rust after C# proved too slow for customers

Technologies: Kubernetes, DB2, C#, Jenkins, .NET Core, COBOL, Kafka, Java, Kotlin, Rust, ELK stack

Senior Software Engineer *Aug 2017 – Jan 2019 (1 year, 5 months)*
Coinify ApS, Copenhagen, Denmark
— *https://coinify.com*

Coinify is a Nordic startup which enables merchants to accept cryptocurrencies in exchange for their goods and services, either in mortar-and-brick or online stores. The company also provides a cryptocurrency trading platform.
Built features across Coinify's customer-facing platform for buying and selling cryptocurrency by credit card and the internal tooling behind it, covering ad-hoc transfers and reconciliation. Co-owned and delivered the 3-D Secure implementation for credit-card payments with one other engineer.

Technologies: Node.js, TypeScript, Docker, AWS, PHP, HTML/CSS

Consultant *Feb 2017 – Aug 2017 (6 months)*
AH Integration Solutions AS, Remote (Brazil – Norway)
— *https://web.archive.org/web/20160208120600/http://ahint.no/*

AH Integrations was a Norwegian company working in the health sector.
Consulted on the design and build of a cost reporting platform, mentoring the intern working alongside me and preparing each day's work in advance to bridge our limited overlap in working hours.

Technologies: C#, WPF, Crystal Reports

CTO *Jun 2015 – Feb 2017 (1 year, 8 months)*
ITOUCH TECH, Sarasota, United States
— *https://itouchtech.us*

ITOUCH TECH is an US-based company which providers FFLs with a digital way to fill ATF4473 forms.
Owned the technology of a four-person company as CTO, from architecture through hands-on delivery. Modernised the touchscreen kiosk that firearms dealers used to complete the federal ATF 4473 form without errors, and that reported its own usage back for per-form billing, automating its updates with Octopus as the fleet grew from one store to fifteen.

Technologies: C#, Windows Forms, WPF, Octopus

IT Specialist – contract *Apr 2014 – Aug 2014 (4 months)*
HBS, Fortaleza, Brazil
— *https://hbs.tv*

Founded in 1999, Host Broadcast Services (HBS) was originally established with the core mandate of being the Host Broadcaster (HB) for the 2002 FIFA World Cup Korea/Japan™; the organisation has since developed as a renowned host broadcast expert.
Ran the IT infrastructure and signal monitoring for HBS's Fortaleza venue at the 2014 FIFA World Cup, supporting roughly 300 venue staff through the pre-tournament build-up and every match hosted there.

Technologies: AD, Windows Server, C#, VB, satellite comm, optical fiber

Senior Software Engineer *Jan 2013 – Mar 2014 (1 year, 2 months)*
Predictive Science, Remote (Brazil – US)
— *https://predsci.com*

Predictive Science Inc. is an employee-owned company that delivers state-of-the-art scientific solutions.
Built the escalation dashboard used by Verizon's Executive Response Team, the group handling escalated, high-profile customer cases, giving them the data to judge which cases genuinely warranted escalation; escalations fell from about 1% to roughly 0.05% once it was in use.

Technologies: Python, Kafka, C#

IT Specialist/Copywriter *Jul 2012 – Dec 2012 (5 months)*
Upstream Systems, Athens, Greece
— *https://www.upstreamsystems.com*

Upstream Systems is a Greek media company that helps mobile carriers to promote products and services to their customers.
Wrote Brazilian Portuguese copy for Brazilian mobile carriers, and built a VBA tool on the team's existing Excel workbook to track copywriting work.

Technologies: Excel, VBA

Senior Software Engineer *Mar 2011 – Dec 2011 (9 months)*
PGE-CE, Fortaleza, Brazil
— *https://pge.ce.gov.br*

Procuradoria Geral do Estado do Ceará (PGE-CE) is the state attorney central for the state of Ceará, in Brazil.
Served as one of the senior developers on the task force that moved the state attorney general's office to digital judicial filings, mandated by Brazil's higher courts, contributing to both the architecture and the implementation while guiding the junior developers on the team. Designed and solely owned PGE-C2, a cloud-based Android app that extended the web-based case management workflow into the attorneys' daily operations.

Technologies: Ruby, Rails, C#, Mono, Postgres, Solr, Android, Java

Senior Software Engineer *Jan 2009 – Feb 2011 (2 years, 1 month)*
SecrelNET, Fortaleza, Brazil
— *https://secrelnet.com.br*

SecrelNET is an ISP of Fortaleza, Brazil. It was one of the pioneers of both dial-up and broadband technologies in the area.
Created Spider, a web tool that let the ISP's round-the-clock first-line support carry out hosting operations, such as creating accounts and sites or restarting a downed IIS site, that had until then required a second-line engineer holding machine-wide credentials. Adopted across most of the company's routine operations, it removed a recurring escalation path and the out-of-hours waits that came with it.

Technologies: .NET, C#, Java, C, Windows Development, WMI, VB, Linux, Hyper-V, Active Directory

Education

Universidade Federal do Ceará, *Fortaleza, Brazil* *2013 – 2016*
— *https://ufc.br*

Bachelor of Science
Economics (Transcript of records)

Universität Liechtenstein, *Vaduz, Liechtenstein* *2015 – 2016*

— *https://uni.li*

Exchange Student
Business Administration and IT (Transcript of records)

Publications

- *Over the Cliff: How Brazil's 2004–2016 Business Cycle Became Its Worst Recession in a Century* (2019), DOI: 10.35297/qjae.010027
- *Understanding Bitcoins: Facts and Questions* (2015), DOI: 10.5935/0034-7140.20150001

Reviews

Bachelor theses:

- *Os efeitos do grau de Liberdade Econômica sobre o Índice de Desenvolvimento Humano (IDH) e o IDH ajustado à desigualdade*, Renato Leitão (2022)
- *Bitcoins - uma moeda global e descentralizada*, Raíssa Guerra (2022)
- *Abertura comercial e produtividade: evidências empíricas de correlação nas últimas duas décadas*, Saulo Queiroz (2021)
- *Mercado de criptoativos: conjuntura e desafios*, Gabriel Radaelly (2021)
- *A moeda digital do século XXI: um estudo de caso sobre o Bitcoin*, Hemerson Braga (2021)

Papers:

- *The Political Thought of José Osvaldo de Meira Penna*, Gabriel F. Benzecry (2023)

Awards

- *SAP DemoJam Barcelona*, 1st (Nov 2015)

Certifications

- *Machine Learning* (2013)
- *Computational Investing, Part I* (2013)

Skills

Programming Languages: C#, Python, SQL, TypeScript, JavaScript, Java, Kotlin, Rust, Ruby, PHP, C, PowerShell, Bash/Shell, VBA, VB, COBOL

AI & Machine Learning: LLM, RAG, Machine Learning, Azure OpenAI, Prompt Engineering, CUDA

Cloud & Containers: Azure, AWS, Kubernetes, OpenShift, Docker, Pulumi, Infrastructure as Code

DevOps & CI/CD: CI/CD, GitOps, Git, GitHub Actions, Azure DevOps, Jenkins, ArgoCD, Octopus Deploy

Data & Event Streaming: Kafka, Kafka Streams, Change Data Capture (CDC), Elasticsearch / ELK, Apache Airflow, PostgreSQL, DB2, Solr

Frameworks & Platforms: .NET / .NET Core, React, Angular, Node.js, Ruby on Rails, WPF, Windows Forms, Android, Mono, HTML/CSS

Systems & Networking: Linux, Windows Server, macOS, Active Directory, Hyper-V, WMI, Networking, Satellite communications, Optical fibre

Other: LaTeX, R / MATLAB / Octave, Crystal Reports

Languages

- Portuguese — Native
- English — Full professional proficiency
- Spanish — Limited professional proficiency
- German — Basic
- Italian — Basic
- Danish — Basic
- Hebrew — Basic

Hobbies

Sports: Go-karts, Martial arts

Music: Singing, Guitar, Piano

Research: Genealogy, History, Mathematics, Physics

Other: Motorbike, Photography, Cooking, Snooker, Hiking